Available online at www.sciencedirect.com
SCIENCE DIRECT®
Explorations in Economic History 43 (2006) 39–63
Explorations in Economic History
www.elsevier.com/locate/eeh

Did banks cause the German industrialization? ☆

Carsten Burhop

University of Münster, Institute for Economic and Social History, Domplatz 20-22, 48143 Munster, Germany

Received 29 July 2003
Available online 16 June 2005

Abstract

In this paper, we discuss the causal relationship between growth of bank assets and economic performance (economic growth, capital accumulation, productivity). We analyze new data for German banking (Burhop, C., 2002. Die Entwicklung der deutschen Aktienkreditbanken von 1848 bis 1913: Quantifizierungsversuche. Bankhistorisches Archiv 28, 103–128.) and improved national accounting data (Burhop, C., Wolff, G.B., 2005. A compromise estimate of Germany's Net National Product 1851–1913 and its relevance for economic growth and cycles. forthcoming, Journal of Economic History.) with several recent VAR/VEC based causality tests. Only weak evidence for a causal influence of banks on economic performance on a nation-wide level is detected. On the other hand, the results support the bank-led growth hypothesis for the modern sector of the German economy. In particular, joint-stock credit banks positively influenced capital formation during the early decades of Germany's industrialization.
© 2005 Elsevier Inc. All rights reserved.

JEL classifications: N 13; N 23; C 32

Keywords: Economic History; Germany; Pre-1913; Financial intermediation; Growth

☆ I thank Maja Micevska, Christian Bayer, Matthias Paustian, Richard Tilly, Guntram Wolff, Seminar participants at the Center for Development Research, and two anonymous referees for many helpful comments and suggestions. All remaining errors are mine. An earlier version of this paper was written while I was a research fellow at the Center for Development Research, Bonn University. Financial support of this institute is gratefully acknowledged.
E-mail address: burhop@uni-muenster.de

0014-4983/$ - see front matter © 2005 Elsevier Inc. All rights reserved.
doi:10.1016/j.eeh.2005.04.005

1. Introduction

The investigation of the banking-growth-nexus is an evergreen in German economic history, in development economics, and in general economics alike. This paper contributes to the ongoing debate by combining a new data set for German banking (Burhop, 2002), improved national accounting data (Burhop and Wolff, 2005), and recent econometric methods (Sims et al., 1990; Toda and Phillips, 1993; Toda and Yamamoto, 1995).

Nineteenth-century Germany is one of the most intensely studied cases of the banking-growth-nexus in economic history (see Fohlin, 1999a; Guinnane, 2002). The idea of a positive impact of large German joint-stock credit banks on economic growth dates back to the writings of Jeidels (1905), Riesser (1910), and Hilferding (1910). It was reintroduced into the debate by Gerschenkron (1962), who hypothesizes that moderately backwarded economies—like Germany during the 19th century—can accelerate their growth by setting-up modern institutions such as joint-stock banks.

This idea was formalized by Da Rin and Hellmann (2002). In their model, banks may act as catalyst for industrialization if they are sufficiently large to provide capital for a critical mass of firms, and if banks possess sufficient market power to make a profit from coordination of industrial activities. In this framework, banks can propel an economy from a self-perpetuating low equilibrium to a sustainable high equilibrium: banks thus can become the driving force in a big push towards industrialization. Banks, for example, can finance a critical mass of firms at below-market interest rates. Firms outside that critical mass are financed at market rates. Banks are interested in financing some firms at lower rates, since profits will be higher after industrialization. They are able to finance some firms at below market interest rates if they have sufficient market power for price discrimination and cost advantages. Such costs advantages are most likely for large banks, since they can realize economies of scale and scope.

Microeconomic theory of banking identifies several ways in which financial intermediaries operate within a world involving information and transaction costs (see Levine, 1997 for an overview). Basically, banks and other financial intermediaries have five fundamental functions: they mobilize savings, allocate resources, exert corporate control, facilitate risk management, and ease the trading of goods, services, and contracts. In addition, banks can successfully screen credit applications, allocating credits only to the most promising investment opportunities, and thereby, fostering technological innovation (King and Levine, 1993a,b). Cost advantages of large banks can emerge from these functions. Large banks may have more screening and monitoring experience since they have more customers and credit applications. They have more power in exerting corporate control since their credits are important for firms. In addition, risk diversification is easier using a large portfolio.

On the other hand, a number of endogenous growth models do not show a causal relationship between financial development and economic growth, e.g., Greenwood and Jovanovic (1990) and Pagano (1993). Pagano, for example, argues that financial development can influence growth by altering the saving rate. Financial development may reduce the saving rate, since consumers have better protection against liquidity

risk and better access to consumer credit. On the other hand, financial development can increase the saving rate, since institutions collecting savings are available to consumers. Furthermore, some authors claim that financial liberalization and financial development can increase the volatility of finance and thereby influence economic growth negatively (Singh, 1997).

Empirically, correlation and causation within the finance-growth nexus is investigated by three types of studies. Time series studies employ the Granger non-causality test (Demetriades and Hussain, 1996) or a modified causality test (Rousseau and Wachtel, 1998; Rousseau and Sylla, 2005; Luintel and Khan, 1999). These studies report mixed results, e.g., a bi-directional causality between indicators of financial development and economic performance. Rousseau and Sylla (2005), employing a tri-variate time-series model, present evidence for a causal link running from financial intermediation to economic growth in the United States between 1790 and 1850. In another paper, Rousseau and Wachtel (1998, p. 672), after investigating the finance-growth nexus for the United States, the United Kingdom, Canada, Sweden, and Norway between 1870 and 1929, concluded that "*In particular, the application of recent time series techniques that use information embedded in the levels of the data indicate clearly that financial development was a driving, causal force behind the rapid industrial transformation experienced by five leading economies prior to the Great Depression*." Furthermore, cross-country studies report positive effects of financial development on economic growth, even after accounting for other factors influencing growth (King and Levine, 1993a,b; Levine and Zervos, 1998; Levine et al., 2000). Finally, studies investigating the financial structure of companies and their sources for finance conclude that the financial system facilitates company growth (Rajan and Zingales, 1998). In a historical context, Becht and Ramirez (2003) show that German mining and steel firms with a close link to joint-stock banks were not liquidity constrained in the early 20th century, whereas firms without such a link were constrained.[1]

For many economic historians, Germany is the pre-eminent example for a bank-led industrialization. Kindleberger (1993, p. 130), concludes that "*the great banks constituted less than a tenth of the total assets of financial institutions of the country but were found at the critical margin affecting economic growth*." Tilly (1986) argues that banks promoted growth through portfolio diversification and the resulting expansion of risk capital. This risk capital could have been used to set up new, large, and innovative enterprises. On the other hand, in a classical and controversially debated paper, Neuburger and Stokes (1974) argue that banks current-account lending negatively influenced German growth.[2] Edwards and Ogilivie (1996) hypothesize that in comparison to the national product joint-stock credit banks were too small to influence the economic development of Germany significantly. Finally, Fohlin (1999b) presents evidence for a modest impact of joint-stock credit banks in Germany on capital formation, capital allocation, and economic growth.

[1] This result contrasts to those of Fohlin (1998), who concluded that credit-bank affiliation did not reduce firms' liquidity constraints.

[2] Debates over their econometric approach ended without a clear-cut result regarding the growth-distortion hypothesis (Fremdling and Tilly, 1976; Komlos, 1978).

A major shortcoming of nearly all quantitative investigations is the focus on the period 1883–1913. So far, data for earlier phases were unavailable. This study employs a new data set for joint-stock credit banking in Germany and thereby quantifies the impact of banks during the early phases of industrialization. Studies employing modern data focus on economy-wide measures of financial intermediation and economic performance, whereas the overwhelming part of the economic-historical literature concerning Germany employs industrial sector and joint-stock credit bank data. We will investigate the two relationships, a economy-wide and a modern-sector link. The data are described in Section 2. The hypotheses are tested using recent econometric methods, which are briefly described in Section 3. Section 4 presents the empirical results, the final Section 5 concludes.

2. Data

The empirical literature, notwithstanding its diverging conclusions regarding the banking-growth nexus, is relatively clear about the variables to be included into the investigation. A standard AK-endogenous growth model includes output per employee, physical capital stock per employee, and the productivity level of the economy. Since we want to evaluate the impact of banks on economic growth, an indicator of financial depth must be included. Thus, monetary capital is added as an additional factor into the production technology.

We investigate two specifications, an economy-wide and a modern-sector specification. In both, we employ three performance and one financial depth series. For the nation-wide data set, the compromise estimate of the German net national product (NNP) is used (Burhop and Wolff, 2005). The nation-wide capital stock data are taken from Hoffmann (1965, p. 253), with corrections for the industrial capital stock (Burhop and Wolff, 2005). The data were transformed to 1913 constant prices using a NDP deflator (Hoffmann, 1965, p. 825) and to per-employee figures (Hoffmann, 1965, p. 205). The productivity level was calculated using the Solow-residuals from a growth accounting exercise, normalizing the productivity level to $1913 = 1$.[3] Growth is accounted for by using the compromise NNP figures, total employment, and the average share of capital income between 1860 and 1913.[4] The financial sector includes joint-stock credit banks, saving banks, credit co-operatives, and private mortgage banks. Data for joint-stock credit banks are from Burhop (2002), data for saving banks from Hoffmann (1965, p. 733), the other data are taken from Deutsche Bundesbank (1976, p. 60). The measure of financial depth employed for the nation-wide specification is total assets of all banks divided by the net national product. The starting point for the nation-wide investigation—the year 1860—is determined by availability of financial intermediation data.

[3] The year of normalization does not influence the results.

[4] The share of capital income was 24.3 percent.

We do not use the traditional national accounting data of Hoffmann (1965) for several reasons. First, the four national accounting series compiled by Hoffmann (1965) and Hoffmann and Müller (1959) have different levels and different cyclical properties. Second, Hoffmann underestimates the industrial capital stock. Third, Hoffmann underestimates capital income in the industrial sector.[5]

The modern sector includes two parts: industry and railways. The three performance measures for the modern-sector study, which starts 1851, are: (1) the value of industrial production (Burhop and Wolff, 2005) and the value of railway production (Hoffmann, 1965, p. 424) in 1913-prices and per employee (Hoffmann, 1965, p. 196); (2) the industrial and railway capital stock (Burhop and Wolff, 2005; Hoffmann, 1965, p. 253) in 1913-prices and per employee; and (3) productivity within the modern sector, calculated from the Solow-residuals from a growth accounting, normalized to $1913 = 1$. Basic data for the growth accounting are the value of industrial and railway production, the industrial and railway employment, the industrial and railway capital stock, and the average capital income share between 1851 and 1913.[6] We measure financial intermediation within the modern sector by the total assets of joint-stock credit banks divided by industrial and railway production. The 1851 starting date for the modern-sector investigation is determined by the availability of national accounting data.

Figs. 1–4 show the evolution of net national product per employee (Y), modern-sector production per employee (Y^M), total capital stock per employee (K), modern-sector capital stock per employee (K^M), nation-wide productivity level (P), modern-sector productivity (P^M), total financial depth (FD), and modern-sector financial depth (FD^M). All of the series show a sustained upward trend. Fig. 1 displays output per employed in the whole economy and for the modern sector. Labor productivity was higher in the modern sector throughout, giving an incentive to move from agricultural to industry.[7] Between 1860 and 1913, labor productivity growth was equal in both sectors.

Fig. 2 shows the capital stock per employed in the total economy and in the modern sector. Until 1875, capital intensity in the modern sector was below the national average. This resulted from a high share of capital in the agricultural sector, especially land and buildings. Between 1860 and 1871, the growth rate of capital stock per employed was slightly higher in the modern sector. Then, up to 1876, capital formation in the modern sector accelerated substantially. This is the well-known *Gründerzeit*. During these years, total output, output per employed, and productivity significantly increased in the modern sector. From the mid-1870s until the turn of the century, capital intensity growth rates were equal in the modern sector and in the total economy. Thereafter, the growth rate in the modern sector was much faster.

[5] See Burhop and Wolff (2005) for an extensive discussion.

[6] The share of capital income in the modern sector was 16.9 percent.

[7] The traditional data of Hoffmann, in contrast, show an industrial labor productivity below net national product per employed until 1907.

Fig. 1. Development of labor productivity, 1851–1913.

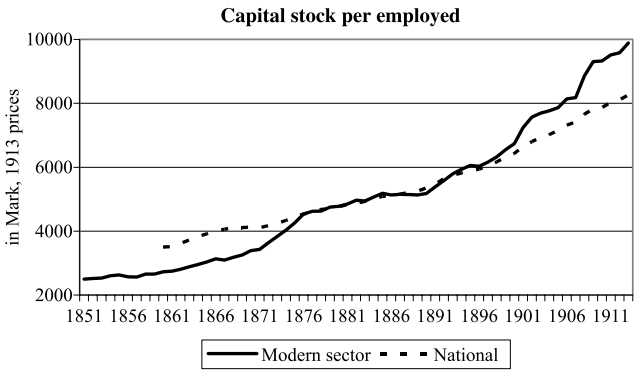

Fig. 2. Capital stock per employed, 1851–1913.

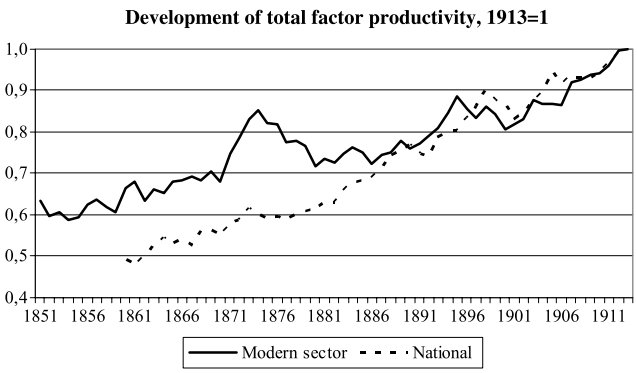

Fig. 3. Development of total factor productivity, 1851–1913. Normalized to 1913 = 1.

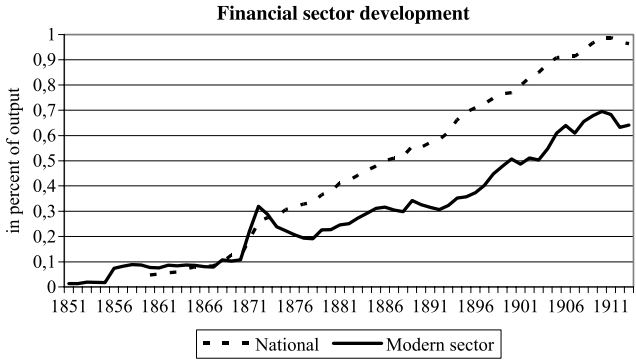

Fig. 4. Financial sector development, 1851–1913.

Fig. 3 presents the development of total factor productivity. From mid-19th century until the late 1880s, total factor productivity growth was stronger in the whole economy than in the modern sector. This is mainly due to the shift of production factors (labor) out of agriculture into high-productivity industry (Broadberry, 1997). The strong upturn of total factor productivity in the modern sector between 1870 and 1874, and the following downturn until 1880 might not be related to increased productivity, but to increased capacity utilization, since the Solow-residual also includes fluctuations of capacity utilization. From the late 1880s until World War I, total factor productivity growth rates were equal in the modern sector and in the whole economy.

Fig. 4 displays financial sector development. The level of financial depth steadily increased, from about 5 percent of NNP (1860) to nearly 100 percent on the eve of World War I. Financial depth related to joint-stock credit banks also rose, namely from 1.3 percent of modern sector output in 1851, to 7.7 percent in 1860. and finally to nearly 65 percent in 1913. Financial intermediation in the whole German economy was greater than within the modern sector. In particular, deregulation brought about by the joint-stock companies act of 1870 led to a sustained increase of joint-stock credit bank and joint-stock mortgage bank activity. This important phase of German economic policy is analyzed in this paper.

There are, however, some shortcomings of financial intermediation data. First of all, private banking houses—like Rothschild, Oppenheim, Bethmann and Bleichröder—are not included into the data set, which leads to an underestimation of financial depth. In particular during the early decades of Germany's industrialization, private credit banks were of outstanding importance. Unfortunately, the quantitative relevance of these banks is uncertain. Tilly (1966) estimates the total assets of private banks in the Rhineland to be around 220 million Marks (1845), 570 million Marks (1855), and 940 million Marks (1865). Goldsmith (1969) estimates the total assets of all German private banks to be around 1.5 billion Marks in 1860 and 2.5 billion Marks in 1880. Private credit banks were thus more important in terms of assets than joint-stock credit banks, at least until around 1890. Thereafter, many

private banks were taken over by joint-stock banks, which led to an extensive growth of the latter.[8] Since the business of the large private banks was quite similar to that of joint-stock credit banks until the turn of the century—in fact many joint-stock credit banks, e.g., Bank für Handel und Industrie, Berliner Handels-Gesellschaft, Deutsche Bank, Dresdner Bank, were founded by private bankers to reduce the personal risk of the fully liable private bankers—we can reasonably assume that our joint-stock credit bank data underestimate the influence of credit banks (joint-stock and private) during the whole period.[9] If we find a positive influence of joint-stock credit banks on economic development, this effect might be stronger if we could have included private banks into our investigation.

Second, the data are aggregated by simply adding up total assets, thereby double counting inter-bank lending, which leads to an overestimation of financial depth. The underestimation of total assets due to leaving out private bankers seems, however, to be far more important than double counting. Finally, up to 1880 the total assets of private mortgage banks were estimated using information about long-term credits of these institutions and the share of long term credits of total assets between 1881 and 1913. This seems reasonable, but it is an assumption.

A further problem could be the sub-sectoral separation. We assume that only joint-stock credit banks supply finance to modern industrial companies, and that these firms are exclusively financed by joint-stock credit banks. This is, off course, a simplification. It is well known that the large joint-stock credit banks were active in other fields. For example, they sold government and foreign bonds on the German capital market, sometimes accompanied by credits. We have no data about the distribution of credits, however, and that is the main reason for our simplification. What we do know is the distribution of joint-stock credit banks assets, which is surprisingly stable over time (Burhop, 2002). Almost in every single year between 1851 and 1913, 40–50 percent of total assets were devoted to current account lending, and about 20 percent to bill discounting. These two activities, which are clearly related to private business, comprise about two thirds of all assets.[10] Until the late 1870s, about 20 percent of all assets were invested in stocks and bonds; thereafter that share fell to about 10 percent. Until the 1870s, almost all bonds were issued by governments and railway companies; the first industrial bond was issued by Krupp in 1874. Stocks were issued by railway companies, banks, and industrial firms. Furthermore, at least until the 1870s, many joint-stock banks had large and long-term investments in some industrial companies.

Another assumption is that savings banks and credit co-operatives supply no finance to the modern sector. This assumption is problematic for the years after

[8] One should note that total assets of private bankers remained stable after 1890, and that the total number of private banks increased.

[9] The similarity of business and sharing of large investment projects is well documented, e.g., by Feldenkirchen (1979, 1982), and Burhop (2004a, Chapter 4).

[10] We do not have much microeconomic evidence on credit distribution. It could be possible to investigate that issues using the general ledgers of banks. This is, however, far beyond the scope of this paper.

about 1890, since by then the saving banks might have started to supply industrial finance. This might be of minor importance, however, since saving banks during the period 1851–1913 invested a relatively constant share—about 55 percent—of their assets in mortgages and about 10 percent of total assets in personal credits. Mortgages were used to finance buildings and distribution of estates (*Erbauseinandersetzung*), neither of which is included in modern-sector investment. Further, saving banks invested around one-third of their assets in government bonds and communal finance. Some of these funds were reinvested in infrastructure construction, which again is not included into industrial investment. Therefore, only a small part of saving banks assets was directed to industrial finance.[11] Yet, if saving banks financed local governments or firms, and these entities bought industrial goods (e.g., for the electrification of cities), savings banks might have had an indirect effect on industrialization. Since total assets of saving banks were larger than those of joint-stock credit banks for most years under consideration—only during 1871–1874 did the reverse hold—we explicitly investigate here the role of saving banks.

3. Econometric methodology

Engle and Granger (1987) show that if two series are individually integrated and cointegrated, a causal relationship will exist in at least one direction. The concept of causality is a statistical one, and it is not related to causality concepts of economic theory. In our context, a time series causes another time series if the former improves the predictive power of a time series model forecasting the second series.

A general test for causality in VARs is the method proposed by Toda and Yamamoto (1995). This test is independent of two properties that macroeconomic time series often have: unit-roots and cointegration. This is of special relevance, since tests for unit-roots and cointegration generally have low power in small samples. Thus, causality tests, depending on unit-root and cointegration properties can well have serious pre-test biases. Moreover, the causality test itself has low power and tends to reject the non-causality hypothesis too often: Zapata and Rambaldi (1997) suggest a rejection rate of about 8 percent on a 5 percent significance level in bi-variate VARs with about 60 observations. Reducing the number of observations or increasing the number of variables reduces the statistical reliability of the results further.

The results of the Toda–Yamamoto causality test can be checked using two additional tests: the traditional Granger-test and the VAR-based test of Sims et al. (1990), which is generalized by Toda and Phillips (1993). The Granger-test is only possible with non-cointegrated, stationary data. On the other hand, the Sims et al. test is only possible with cointegrated data. The test uses the fact that a causal relationship between cointegrated series must exist. Furthermore, the test developed by

[11] Wysocki (1993) discusses the role of saving banks for industrialization and summarizes the relevant data.

Toda and Yamamoto cannot be used if the maximum number of unit-roots in the VAR is larger than the optimal lag-length. Thus, in some cases, it might not be possible to conduct causality tests.

Some pre-tests are therefore necessary to choose the correct types of non-causality tests. First, the order of integration must be tested. For this part, we employ the augmented Dickey–Fuller (ADF) and the Phillips–Perron unit-root tests. Then bi-variate cointegration tests using Johansen's approach are conducted for all integrated variables. If bi-variate cointegration exists, then uni- or bi-directional causality will exist, although in finite samples there is no guarantee that causality tests will identify it. In a third step, multivariate cointegration tests are performed to examine interaction effects. Finally, causality tests can be performed.

The starting point is the Toda–Yamamoto test. To perform this test, we need to know the maximum order of integration involved in the model, $d_{\max}$, and the optimal lag-length for a VAR.[12] We can then estimate a VAR in levels with a lag length of optimal lag length plus $d_{\max}$. Causality is inferred only from the optimal lag-length VAR coefficients using standard Wald- or t-tests. In the bi-variate case:

$$X_t = \alpha + \sum_{i=1}^{m+d_{\max}} \beta_i X_{t-i} + \sum_{j}^{n+d_{\max}} \gamma_j Y_{t-j} + u_t, \tag{1}$$

$$Y_t = a + \sum_{i=1}^{q+d_{\max}} b_i Y_{t-i} + \sum_{j}^{r+d_{\max}} c_j X_{t-j} + v_t, \tag{2}$$

where u_t and v_t are zero-mean, serially uncorrelated, random disturbances and m, n, q, and r are the optimal lag-lengths.

In case of sufficiently cointegrated data, we may also use the level-series and estimate a VAR, using only the optimal lag-lengths m, n, q, r. This method is proposed by Sims et al. (1990) and generalized by Toda and Phillips (1993). Finally, if the data are $I(1)$ but not cointegrated, we employ the first differences of the original series and the standard Granger-test. We can formulate the non-causality hypothesis

$H_0 : Y$ does not cause X if $\gamma_j = 0 \, \forall \, j$ is not rejected.

4. Empirical results

4.1. Results of causality tests for joint-stock banks and all banks

To perform causality-tests in time series models, it is crucial to know, whether the data are integrated and cointegrated. The ADF unit-root test with trend and intercept and the corresponding Phillips–Perron unit-root test indicate that all time series are $I(1)$.

[12] The optimal lag-length is calculated using the Hannan–Quinn criteria.

Johansen's cointegration test with intercept and trend is employed to estimate the number of cointegration relationships. In all cases, the trace test and the maximum eigenvalue test indicate the same number of cointegration relationships. The following series combinations were not cointegrated in either the country-wide or the modern-sector specifications: the tri-variate system including financial intermediation, income per employed, and productivity level, and the two bi-variate systems with financial intermediation combined with income per employed or productivity. The classical Granger-causality test thus is only possible for this three combinations of series; in all other cases the test developed by Sims, Stock, and Watson can be used.

In all cases, the optimal lag-length was at least as large as the maximum number of unit-roots in the system. Therefore, the test developed by Toda and Yamamoto can be applied to all combinations. One should note that this test has a lower statistical power than the classical Granger-test and the Sims–Stock–Watson test since more parameters are estimated.

Table 1 shows the *p*-values of the non-causality tests for two periods, 1860–1913 and 1883–1913. Table 2 displays the results for the modern sector. Here three periods (1851–1913, 1851–1882, and 1883–1913) are distinguished.

There is no historical reason to split the sample in 1883. The underlying nation-wide time-series do not have a structural break. The modern sector series might have structural breaks in 1870–1871 and around 1880. Earlier macroeconomic studies, however, rely on a data set starting in 1883; taking 1883 as a break-year therefore

Table 1
Non-causality test results for the nation-wide specification

Included variables	Toda–Yamamoto		Granger		Sims et al.	
	H0: FD did not cause *p*-value	H0: FD is not caused by *p*-value	H0: FD did not cause *p*-value	H0: FD is not caused by *p*-value	H0: FD did not cause *p*-value	H0: FD is not caused by *p*-value
Sample period: 1860–1913						
Y	0.152	0.813			0.810	0.828
K	0.305	0.918			0.427	0.660
P	0.398	0.906			**0.029**	0.876
K	0.364	0.934			0.653	0.291
P	0.616	0.751			*0.072*	0.973
Y	0.131	0.653			0.430	0.875
K	0.453	0.926			0.558	0.527
Y	0.137	0.783	0.132	0.710		
P	0.376	0.875	0.724	0.998		
Y	0.104	0.659	0.134	0.635		
P	0.633	0.799	0.748	0.773		
K	0.424	0.987			0.426	0.257
Sample period: 1883–1913						
Y	0.460	0.193	0.763	0.110		
P	**0.005**	0.573	0.771	0.373		
K	**0.000**	**0.013**	*0.057*	0.665		

Bold fonts indicate significance on 5 percent level, italic fonts indicate significance on 10 percent level.

Table 2
Non-causality test results for the modern sector specification

Included variables	Toda–Yamamoto		Granger		Sims et al.	
	H0: FD^M did not cause *p*-value	H0: FD^M is not caused by *p*-value	H0: FD^M did not cause *p*-value	H0: FD^M is not caused by *p*-value	H0: FD^M did not cause *p*-value	H0: FD^M is not caused by *p*-value
Sample period: 1851–1913						
Y^M	0.117	**0.024**			**0.015**	0.217
K^M	0.631	0.304			**0.018**	0.504
P^M	0.110	**0.033**			**0.033**	0.206
K^M	0.641	0.790			**0.016**	0.104
P^M	0.212	0.447			**0.037**	0.719
Y^M	0.192	0.272			**0.015**	0.807
K^M	0.629	0.854			**0.016**	0.125
Y^M	0.130	**0.026**	0.249	0.329		
P^M	0.231	**0.040**	0.282	0.397		
Y^M	0.158	0.382	0.281	0.727		
P^M	0.211	0.703	0.265	0.861		
K^M	**0.002**	0.128			**0.003**	**0.007**
Sample period: 1851–1882						
Y^M	**0.013**	0.481	*0.060*	0.511		
P^M	**0.023**	0.727	**0.046**	0.541		
K^M	**0.000**	0.368	**0.000**	0.554		
Sample period: 1883–1913						
Y^M	0.769	0.184	0.481	0.308		
P^M	0.765	0.269	0.460	0.442		
K^M			0.317	0.729		

Bold fonts indicate significance on 5 percent level.

makes our results directly comparable to previous research. Furthermore, taking 1883 as a breakpoint leaves 31 observations in the latter sample and 32 observations in the earlier sample. This is an advantage, since estimation results are certainly unreliable with fewer than 30 observations.

Financial sector developments were not a causal force for output, productivity, and capital formation in the whole German economy between 1860 and 1913. The Toda–Yamamoto test fails to reject the non-causality hypothesis in all investigated cases. This result is largely confirmed by the classical Granger and the Sims–Stock–Watson test. There is only weak evidence that financial sector development caused aggregate productivity during these decades. One should note that aggregate productivity on the nation-wide level was not mainly driven by technological progress, but by structural change. Thus, financial intermediaries might have had causal influence on structural change.

During the later period, 1883–1913, the results are more favorable to a causal role for the financial sector. The Granger test rejects the null hypothesis of no causal influence of the growth of the financial sector on net investment at a 10 percent significance level. The Toda–Yamamoto test, however, indicated that the levels of

financial sector development and the capital stock are endogenous. But this test rejects the null hypothesis of no causal relationship between the level of financial intermediation and the productivity level of the whole economy.

Jointly and severally, the impact of the level of financial-sector development and financial-sector growth does not appear to have had a causal influence on Germany's economic development during the second half of the 19th and the early 20th century.

Turning to the joint-stock credit banks, the econometric results are more favorable to a causal role for finance (see Table 2).

For the whole period 1851–1913, the evidence is somewhat ambiguous: the Sims–Stock–Watson test supports the hypothesis of a leading role of the joint-stock credit banks, whereas the less efficient Toda–Yamamoto test gives some support for the reverse case. However, the efficiency of causality tests in small samples with more than two variables is rather low. If we focus on the results of the bi-variate VARs, the Toda–Yamamoto test rejects the null hypothesis of no causal relationship between the level of financial intermediation and the level of capital stock per employed in the modern sector, whereas the test by Sims, Stock, and Watson supports a bi-directional causality.

Interpretation of results is more straightforward after the sample is split. For the later period 1883–1913, no causal relationship between joint-stock credit banks and economic performance is detected. In contrast, joint-stock credit banks were a causal force for economic development during the early phase 1851–1882. The Toda–Yamamoto test rejects the hypothesis of no causal relationship between the level of financial intermediation by the joint-stock credit banks and the level of output per employed, capital stock per employed, and productivity. This result is confirmed by the traditional Granger test, which rejects the null hypothesis of no causal relationship between the change of financial intermediation via the joint-stock credit banks and the growth of output, capital stock, and productivity.

The joint-stock credit banks seemingly lost their special influence after 1882: the non-causality hypothesis cannot be rejected. This evidence might support Da Rin and Hellmann's model, which concludes that banks can induce a shift from a low- to a high-equilibrium growth path. Our results indicate that German joint-stock credit banks were causal forces in the transition phase, but not a causal force when the economy had already reached a high-equilibrium growth path. One can thus argue that the German economy was on a sustained industrial growth path after the 1870s. Our results also support recent research by Edwards and Ogilivie (1996) and Fohlin (1999b), who contend that the influence of joint-stock credit banks during the period 1883–1913 was small at best. They can be taken as well to support Gerschenkron's argument that the role of credit banks was particularly important in the early stages of German industrialization when the economy was relatively backward.

Having established an early causal relationship between banks and modern-sector growth, we can investigate the importance of joint-stock credit banks for capital formation, output and productivity growth in the modern sector. We address this question by employing impulse–response techniques in VARs covering the years 1851–1882. This impulse–response analysis shows how the components of a VAR

dynamically react to a one-standard-deviation shock to modern sector financial intermediation. Figs. 5–7 show the responses of the first difference of output per employee, capital stock per employee, and productivity in the modern sector during 1851–1882 on a shock to financial intermediation by the joint-stock credit banks. The figures include the point-estimate and a one standard deviation confidence interval.

In all three cases, a financial intermediation shock has a positive influence on the performance measures. The cumulated effects over 10 years are (in percent of the sample mean of the variable) 320 percent for output per employee, 389 percent for productivity, and 204 percent for capital per employee. Another measure of the influence of a shock is the share of explained variance. The financial intermediation series explains 13 percent of the variance of the productivity series after a financial shock. The corresponding figure for labor productivity is 14 percent, and finally for capital intensity, 40–60 percent from the third year onwards. Thus, fluctuations of financial intermediation strongly influence fluctuations of net investment. Analyzing the response over time yields the interesting result that output and productivity quickly

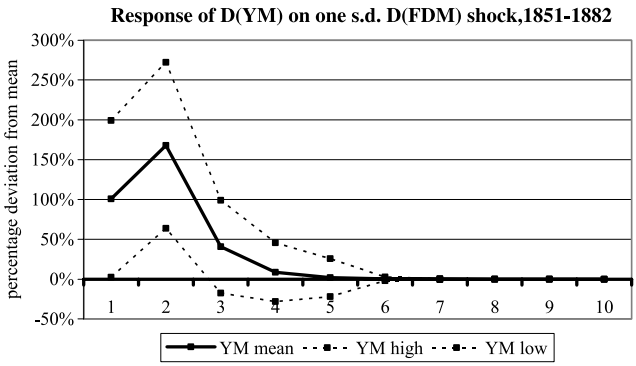

Fig. 5. Response of $D(Y^M)$ on one standard deviation $D(FD^M)$ shock, 1851–1882.

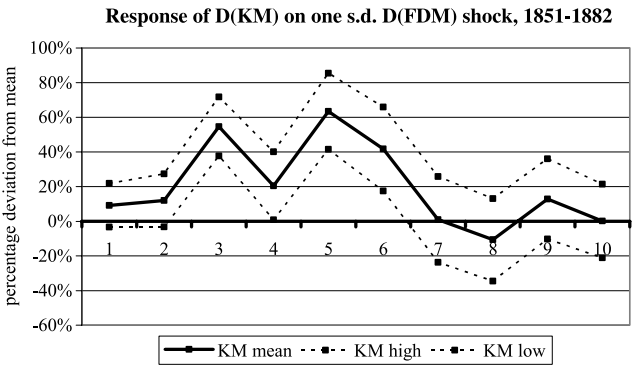

Fig. 6. Response of $D(K^M)$ on one standard deviation $D(FD^M)$ shock, 1851–1882.

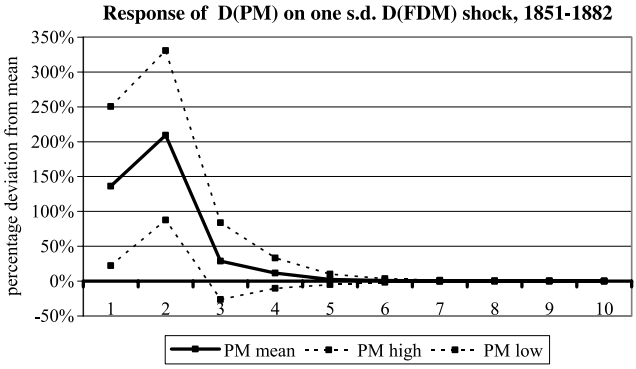

Fig. 7. Response of $D(P^M)$ on one standard deviation $D(FD^M)$ shock, 1851–1882.

react to a financial shock, whereas capital formation reacts with some delay. Output and productivity both react in the first period after the shock. They then return to their equilibrium growth path. Since it is unlikely that new technologies, which were financed by the banks, would have an immediate influence on productivity, it seems that the measured influence is related to capacity utilization. Variations in capacity utilization are included into the Solow-residual. An assumption of growth accounting is full employment of all production factors. In practice, however, factor utilization is less than 100 percent. The time structure of the response of net investment (first difference of the capital stock) is more complicated. Net investment increased in the second year after the financial shock, and it remained above its equilibrium level until the fifth period after the shock. Thus, a financial shock positively influenced capital formation for about 4 years.

4.2. Influence of other types of banks

One can question a special role for joint-stock credit banks if other types of financial intermediaries were also a causal force for income growth and capital formation. Candidates for such a role are private bankers and saving banks. Unfortunately, we cannot further investigate the role of private bankers on a macroeconomic level since time series of total assets or asset structure are not available.[13] However, a qualitative assessment of private banker's role is possible.[14] Private bankers were the first financing large-scale private investment, especially railway construction during the 1830s and 1840s. Important in this regard were bankers from Cologne and the sur-

[13] It would be possible to construct such a series employing Goldsmith's point estimates for 1860, 1880, 1900, and 1913, linking them with a log-linear trend, and adding a business cycle component using trend-cycle decomposition of other banks assets. However, the historical base of such a time series would be rather thin.

[14] Reitmayer (2002) and Wixforth and Ziegler (1994) discuss the role and influence of private bankers during the late 19th and early 20th century.

rounding Rhineland, since they actively started railway finance. Their fellow private bankers from Frankfurt and Berlin were mostly active in government finance, a business not important in Cologne, since there was no seat of government and no close link to foreign governments. Private bankers involved in railway finance realized the high risk involved, and intended to found joint-stock banks to reduce personal risk. The Prussian government for political reasons refused to found a joint-stock bank until 1848. From 1848 until the mid-1850s, several large credit banks were created, e.g., Schaaffhausen (1848), Bank für Handel und Industrie (1853), Disconto-Gesellschaft (1853), and Berliner Handels-Gesellschaft (1856). Three of them were created with large capital shares from Cologne's bankers and the fourth (Disconto) was managed by a Rhenisch businessman. Private bankers with experience in financing large-scale private investment thus founded joint-stock banks to reduce their personal risk and to institutionalize financing consortia. Until the 1870s, these joint-stock credit banks were not independent of their private-bank founders, who took up positions in management and whose own firms participated in most projects. After the liberalization of the joint-stock companies act in 1870, many new banks were formed. Again, private bankers played a leading role in this process. Until the mid-1870s, private bankers controlled the joint-stock banks, and the banking business was highly personalized: private bankers, bank managers, and businessmen from industry knew each other and were well informed about business.[15] After the 1873 stock market crisis, conflicts of interest became obvious for managers and shareholders of joint-stock banks: private bankers had an incentive to transfer risks from their private business to joint-stock banks, thereby reducing the return for shareholders and the income of managers.[16] During the late 1870s and 1880s, managers of joint-stock banks separated their businesses from private bankers' influence. At the same time, personalized information networks became insufficient within a complex economic-industrial environment. Thus, private bankers lost much of their information advantage over the formalized organization of large joint-stock banks. Private bankers remained an influential group within Germany, however, especially for government finance and as independent consultants for industrial firms.

Saving banks were another important type of bank. For savings banks, time series of total assets and asset structure are readily available. They have higher total assets than joint-stock credit banks for most years between 1851 and 1913 (see Fig. 8). Total assets of these two types of banks closely co-moved until the mid 1870s, and then again from the mid 1890s until World War I. During the two decades from about 1875 to 1895, total assets of joint-stock credit banks stagnated, perhaps as a result of the aftermath of the 1873 stock market crises.

For the econometric evaluation, total assets of savings banks were related to developments in the whole economy and in the modern sector. Relating savings bank assets to performance measures for the whole economy leads to no clear-cut results.

[15] Da Rin (1996) describes the different phases of bank information collection during the period 1850–1913.

[16] Burhop (2004b) shows that about three quarters of bank managers income were profit shares.

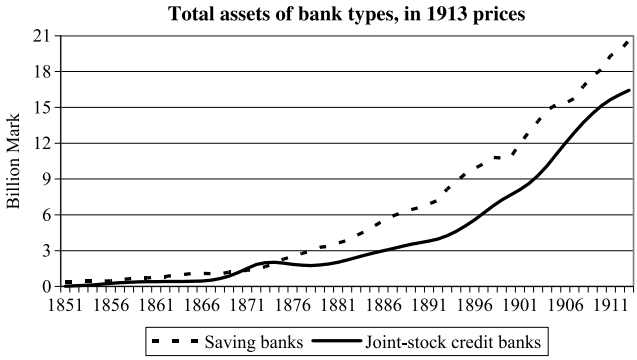

Fig. 8. Total assets of saving banks and joint-stock credit banks, in 1913 billion Mark.

All variables were endogenous between 1851 and 1913, as well as during the two sub-periods 1851–1882 and 1883–1913.

More interesting are the findings for the modern-sector specification. Between 1851 and 1913, a causal influence of the saving banks on capital formation is detected. This finding is supported by the Toda–Yamamoto test for a causal relationship between the level of financial intermediation and capital intensity, and by the classical Granger-test for the causal relationship between the first differences of financial intermediation and capital intensity.

The causal relationship is confined to the second sub-period only. Between 1883 and 1913, both tests—the Granger-test and the Toda–Yamamoto test—reject the null hypothesis of no causal relationship between savings bank financial intermediation and capital formation. But for the earlier sub-sample covering the years 1851–1882, the null hypothesis of no causal relationship is not rejected.[17] Therefore, the savings banks had a causal influence on capital formation in Germany during the later stages of economic development, but not during the "take-off" phase.

4.3. Historical and theoretical interpretation

In this section, we connect our empirical results to the model of Da Rin and Hellmann (2002) and we support our statistical results with historical evidence. Da Rin and Hellmann (2002, Proposition 2) show that an economy can move from a low-equilibrium (no industrialization) to a high equilibrium (industrialization) with the help of banks if some banks have sufficient market power to supply finance to a critical mass of firms at below-market interest rates. By financing some firms at low costs, these firms become willing to invest in new, high-risk projects, irrespective of whether firms outside the critical mass also invest. By definition, the critical mass

[17] It that case, only the Toda–Yamamoto test is possible. For the Granger-test, the optimal lag-length is so large that the VAR is not identified. Furthermore, the Sims–Stock–Watson test is in no case applicable since the data are not cointegrated.

is large enough to induce a shift to the high equilibrium. In the high equilibrium, all firms are willing to invest and interest rate discrimination by large banks is not necessary anymore.

In this regard, market power has two dimensions: powerful banks must have a cost advantage over other banks to supply some firms with low-cost capital. In addition, powerful banks must be large enough to supply the critical mass of firms with capital. Quantitative evidence for Germany fits well with these two dimensions: the Herfindahl-Index for the joint-stock credit banks was around 0.1 between the late 1850s and the early 1870s. After the liberalization of the joint-stock companies act in 1870, it falls to 0.02. The market share of the three largest joint-stock credit banks in the joint-stock credit bank market was larger than 50 percent until 1870; thereafter, this share fell to less than 30 percent (Burhop, 2002). Furthermore, we have evidence for a concentration of joint-stock credit bank activity: these banks relied heavily on business relations with few large customers, at least during the 1870s (Burhop, 2004a). Thus, we can expect low monitoring costs for bank managers.

There are several reasons for finding a stronger link between modern-sector capital formation and joint-stock credit banks than for the nation-wide specification. First, many of the *Kreditbanken* were explicitly founded to finance industrial development, e.g., the Bank für Handel und Industrie in 1853, and the Berliner Handels-Gesellschaft in 1856. Already in the 1830s, private bankers financed the first German railroads. They organized the initial public offerings (IPOs), they supplied current account credits to the newly established companies, and they supervised the development of the business by taking up board memberships. These actions were not done by single private bankers, but by syndicates of private bankers. Thereby, they allocated the risk over several private banking houses and collected capital from all over Germany. These IPO syndicates were the starting point for the later joint-stock credit banks, since many of these were founded by private bankers. Initially, the joint-stock banks were seen as a way to reduce the personal risk of large IPOs.[18]

From the early 1850s onwards, a division of labor between short- and long-term investments was in fact effective. Short-term credits, bill discounting or issuing of bank notes for example, were arranged by banks of issue. In Prussia, the Prussian Bank fulfilled this role beginning in late 1840s. Sometimes, the division of labor was made explicit by the founding of separate institutes for short- and long-term credits. The first example in this regard is the founding of the Bank für Handel und Industrie (for long-term investments) and of the Süddeutsche Bank (a private bank of issue for short-term credits) by the same founders in 1853 and 1856. The two banks were located in the same building and they were managed by the same board. Thus, the joint-stock credit banks could focus their business on long-term industrial finance.

Banks with long-term commitments to some industrial firms could successfully screen companies and investment projects, and finance only those firms and projects using the most promising new or imported technologies. In this regard interlocking

[18] See Burhop (2004a) for an overview of joint-stock credit banking in Germany.

directorates between the large banks and industrial companies could be important.[19] Fohlin (1999c) showed that this relationship became important on a nation-wide level only after 1895. Nevertheless, before that date, we have evidence for interlocking directorates between banks and industrial firms, at least for those firms doing stock-exchange business with the banks (Burhop, 2004a; Feldenkirchen, 1979, 1982).

Moreover, the joint-stock credit banks were large banks with (at least after 1890) nationwide operations. This facilitated risk-pooling, since many firms and many regions were debtors (and creditors) of these banks. Since the banks operated in different sectors of the economy and worked on a national basis, they successfully exploited differences in risk preferences between sectors and regions. They reduced diversifiable risks by allocating their credits over a larger geographic area and over customers from different sectors.

Joint-stock credit banks did not accomplish all functions from the beginning. The main reason for their foundation was risk reduction. From the outset, they screened credit applications and built up long-term relationships with industrial customers. National and international payment systems were established during the 1850s, for example, by the Bank für Handel und Industrie via a large network of *Kommanditbeteiligungen*. The large credit banks spread their credits over many regions and sectors, during the early years mostly via credit or IPO syndicates. Later on, branches in other cities were established; e.g, the Bank für Handel und Industrie expanded from Darmstadt to Frankfurt during the mid 1860s, and to Berlin in 1871. The Deutsche Bank, founded 1870 in Berlin, expanded to Bremen in 1871 and Hamburg in 1872.

A major early weakness of the joint-stock credit banks was the collection of savings. This activity accelerated during the 1890s after the systematic set-up of deposit offices by the large joint-stock banks. Savings collection was much better achieved by the savings banks and credit co-operatives. But for them, the allocation of this capital was tightly restricted; for example, stock-market business was prohibited for saving banks. Moreover, savings banks had to invest their funds in safe investments such as government or communal bonds. Furthermore, the interregional allocation of capital was suboptimal, since the funds of credit co-operatives and saving banks were to a large extent locally bounded. This localization of credits and deposits also increased the risk for these banks, since regional and sectoral economic shocks could not be balanced by developments in other regions or sectors. On the other hand, the co-operatives and saving banks were quite successful in screening credit applications and monitoring creditors, since bankers and creditors belonged to the same community (Guinnane, 2001, 2003). Altogether, the joint-stock credit banks fulfilled four of five fundamental functions of financial intermediaries—allocation of credits, exertion of corporate control, facilitation of risk management, and transfer of payments—better than other financial intermediaries. Other financial intermediaries, especially saving banks, had advantages in collecting savings.

For the later decades of the 19th and the early 20th century, however, a causal influence of joint-stock credit banks on income growth, productivity, and capital

[19] See Fohlin (1999b,c) regarding interlocking directorates.

formation is not supported by our statistical analysis. A possible reason for this could be higher monitoring costs as the German economy grew. Baliga and Polak (2004) argue that in economies with strong bargaining power of banks against other firms and with relatively low private wealth, monitored bank loans are superior to non-monitored tradable debt contracts like bonds. When the monitoring costs are high because there are many firms, a market based financial system Pareto-dominates a bank-based one. But when monitoring costs are low, for example, when banks make loans to a few big customers, a Pareto-ranking of market based vs. bank based financial systems is not possible. Problematic is the stability of a bank-based financial system regardless of its efficiency: thick financial markets must be developed to give industrial firms alternative finance opportunities. This can be done by banks via securitization of loans, for example, or by industrial firms, but only if banks and firms coordinate participation in financial markets. Banks could impede the development of thick financial markets if they do not securitize their loan portfolios. Thus, a once-efficient financial system, which had been a driving force in economic development, could become inefficient within a different information and economic environment.

5. Conclusion

In this paper, we analyze the causal influence of financial sector development on economic performance in 19th century Germany. We use a comprehensive data set for Germany, covering the years 1860–1913, and a new data set for joint-stock credit banks covering the years 1851–1913 (Burhop, 2002). These long time series make use of recent econometric causality tests possible.

Our main result supports the hypothesis of a vital role of joint-stock credit banks for the early industrial development of 19th-century Germany. Total assets of credit banks positively influenced capital formation in the industrial sector between 1851 and 1882. On the other hand, using economy-wide data for financial depth, national income, capital stock, and productivity, we detect no leading role of the financial sector during 1860–1913. During the later stages (1883–1913), savings banks rather than credit banks were more of a causal force for capital formation. The results can be taken as evidence in favor of Gerschenkron's contention that the modernizing role of credit banks was greatest in the early phases of Germany's industrialization when its economy may have been relatively backward.

Future research could focus on three areas: First, since the power of causality tests significantly increases with the number of observations, a forward extension of the data series could yield better results. Furthermore, an international comparative study could investigate whether banks played a special role in the German industrialization, or whether the role of banks was identical in different countries. Finally, a microeconomic study using the general ledgers of banks could yield important insights into asset distribution of joint-stock banks.

Appendix A

Nation-wide					Modern-sector			
Year	Financial depth	Income per employee, in Mark, 1913-prices	Capital stock per employee, in Mark, 1913-prices	Productivity level	Financial depth	Income per employee, in Mark, 1913-prices	Capital stock per employee, in Mark, 1913-prices	Productivity level
1851					0.013	1161	2506	0.63
1852					0.013	1099	2523	0.60
1853					0.020	1116	2533	0.61
1854					0.019	1090	2610	0.59
1855					0.018	1101	2636	0.59
1856					0.074	1154	2575	0.63
1857					0.082	1172	2565	0.64
1858					0.089	1146	2663	0.62
1859					0.087	1127	2658	0.61
1860	0.048	928	3.505	0.49	0.077	1235	2730	0.66
1861	0.052	911	3.524	0.48	0.076	1268	2752	0.68
1862	0.056	926	3.615	0.50	0.086	1186	2807	0.63
1863	0.059	975	3.724	0.52	0.084	1244	2892	0.66
1864	0.073	1.012	3.821	0.54	0.087	1234	2962	0.65
1865	0.081	1.028	3.912	0.53	0.085	1288	3036	0.68
1866	0.081	1.024	3.999	0.54	0.081	1299	3135	0.68
1867	0.084	993	4.058	0.53	0.079	1316	3100	0.69
1868	0.097	1.011	4.116	0.56	0.108	1303	3182	0.68
1869	0.127	1.026	4.113	0.56	0.103	1352	3252	0.71
1870	0.124	1.014	4.129	0.55	0.108	1315	3395	0.68
1871	0.187	1.007	4.108	0.58	0.221	1445	3430	0.75
1872	0.257	1.010	4.176	0.59	0.320	1529	3639	0.78

(*continued on next page*)

Appendix A (*continued*)

Nation-wide					Modern-sector			
Year	Financial depth	Income per employee, in Mark, 1913-prices	Capital stock per employee, in Mark, 1913-prices	Productivity level	Financial depth	Income per employee, in Mark, 1913-prices	Capital stock per employee, in Mark, 1913-prices	Productivity level
1873	0.274	1.017	4.234	0.61	0.287	1631	3835	0.83
1874	0.277	1.020	4.336	0.60	0.238	1689	4030	0.85
1875	0.306	1.042	4.425	0.59	0.223	1646	4263	0.82
1876	0.317	1.044	4.535	0.60	0.207	1655	4534	0.82
1877	0.329	1.042	4.602	0.59	0.194	1576	4620	0.78
1878	0.336	1.078	4.682	0.60	0.191	1582	4627	0.78
1879	0.365	1.087	4.708	0.61	0.227	1564	4750	0.77
1880	0.377	1.056	4.761	0.61	0.227	1467	4778	0.72
1881	0.412	1.088	4.829	0.63	0.246	1507	4850	0.74
1882	0.422	1.093	4.873	0.63	0.251	1494	4966	0.73
1883	0.442	1.126	4.930	0.66	0.273	1535	4941	0.75
1884	0.462	1.165	4.991	0.68	0.292	1574	5064	0.76
1885	0.479	1.193	5.086	0.68	0.311	1552	5184	0.75
1886	0.500	1.199	5.110	0.69	0.316	1499	5131	0.72
1887	0.511	1.215	5.160	0.71	0.305	1541	5151	0.74
1888	0.518	1.245	5.203	0.74	0.298	1555	5146	0.75
1889	0.557	1.243	5.248	0.75	0.342	1610	5129	0.78
1890	0.554	1.237	5.349	0.77	0.326	1573	5174	0.76

1891	0.572	1.241	5.467	0.75	0.315	1610	5386	0.77
1892	0.584	1.258	5.648	0.74	0.306	1655	5585	0.79
1893	0.612	1.336	5.764	0.79	0.323	1704	5796	0.81
1894	0.662	1.362	5.822	0.80	0.352	1793	5943	0.85
1895	0.693	1.381	5.898	0.80	0.357	1879	6059	0.89
1896	0.710	1.399	5.940	0.83	0.374	1815	6030	0.85
1897	0.725	1.424	6.043	0.86	0.402	1781	6160	0.83
1898	0.748	1.472	6.173	0.90	0.448	1843	6324	0.86
1899	0.765	1.454	6.321	0.88	0.479	1820	6548	0.84
1900	0.770	1.437	6.430	0.86	0.508	1746	6738	0.81
1901	0.796	1.448	6.644	0.83	0.487	1798	7240	0.82
1902	0.830	1.466	6.804	0.85	0.511	1837	7569	0.83
1903	0.849	1.503	6.921	0.87	0.503	1940	7696	0.88
1904	0.882	1.530	7.017	0.90	0.548	1927	7762	0.87
1905	0.909	1.578	7.153	0.94	0.610	1929	7864	0.87
1906	0.913	1.562	7.316	0.92	0.640	1936	8138	0.87
1907	0.913	1.574	7.414	0.93	0.609	2060	8178	0.92
1908	0.938	1.638	7.659	0.93	0.655	2101	8857	0.93
1909	0.967	1.641	7.821	0.93	0.678	2149	9308	0.94
1910	0.989	1.634	7.868	0.94	0.696	2155	9325	0.94
1911	0.988	1.682	8.018	0.97	0.683	2201	9519	0.96
1912	0.970	1.688	8.111	0.99	0.633	2289	9579	1,00
1913	0.964	1.734	8.265	1,00	0.641	2306	9892	1,00

References

Baliga, S., Polak, B., 2004. The emergence and persistence of the Anglo-Saxon and German financial systems. Review of Financial Studies 17, 129–163.

Becht, M., Ramirez, C.D., 2003. Does bank affiliation mitigate liquidity constraints? Evidence from Germany's universal banks in the pre-World War I period. Southern Economic Journal 70, 254–272.

Broadberry, S.N., 1997. The Productivity Race: British Manufacturing in International Perspective, 1850-1990. Cambridge University Press, Cambridge.

Burhop, C., 2002. Die Entwicklung der deutschen Aktienkreditbanken von 1848 bis 1913: Quantifizierungsversuche. Bankhistorisches Archiv 28, 103–128.

Burhop, C., 2004a. Die Kreditbanken in Der Gründerzeit. Steiner Verlag, Stuttgart.

Burhop, C., 2004b. Executive compensation and firm performance: the case of large German banks, 1854–1910. Business History 46, 525–543.

Burhop, C., Wolff, G.B., 2005. A compromise estimate of Germany's Net National Product 1851–1913 and its relevance for economic growth and cycles. Journal of Economic History, forthcoming

Da Rin, M., 1996. Understanding the development of the German Kreditbanken, 1850–1914: an approach from the economics of information. Financial History Review 3, 29–47.

Da Rin, M., Hellmann, T., 2002. Banks as catalysts for industrialization. Journal of Financial Intermediation 11, 366–397.

Demetriades, P.O., Hussain, K.A., 1996. Does financial development cause economic growth? Time-series evidence from 16 countries. Journal of Development Economics 51, 387–411.

Deutsche Bundesbank, 1976. Deutsches Geld- und Bankwesen in Zahlen 1876–1976. Fritz Knapp Verlag, Frankfurt am Main

Edwards, J., Ogilivie, S., 1996. Universal banks and German industrialisation: a reappraisal. Economic History Review 49, 427–446.

Engle, R.F., Granger, C.W., 1987. Cointegration and error correction presentation: representation, estimation, testing. Econometrica 55, 1057–1072.

Feldenkirchen, W., 1979. Banken und Stahlindustrie im Ruhrgebiet. Bankhistorisches Archiv 5, 26–52.

Feldenkirchen, W., 1982. Kölner Banken und die Entwicklung des Ruhrgebiets. Zeitschrift für Unternehmensgeschichte 27, 81–106.

Fohlin, C., 1998. Relationship banking, liquidity, and the German industrialization. Journal of Finance 53, 1737–1758.

Fohlin, C., 1999a. Universal banking in pre-World War I Germany: model or myth?. Explorations in Economic History 36 305–343.

Fohlin, C., 1999b. Capital mobilisation and utilisation in latecomer economies: Germany and Italy compared. European Review of Economic History 2, 139–174.

Fohlin, C., 1999c. The rise of interlocking directorates in imperial Germany. Economic History Review 52, 307–333.

Fremdling, R., Tilly, R.H., 1976. German banks, German growth and econometric history. Journal of Economic History 36, 416–424.

Gerschenkron, A., 1962. Economic Backwardness in Historical Perspective. Harvard University Press, Cambridge, MA.

Goldsmith, R.W., 1969. Financial Structure and Development. Yale University Press, New Haven.

Greenwood, J., Jovanovic, B., 1990. Financial development, growth and the distribution of income. Journal of Political Economy 98, 1076–1107.

Guinnane, T.W., 2001. Cooperatives as information machines: German rural credit cooperatives, 1883–1914. Journal of Economic History 61, 366–389.

Guinnane, T.W., 2002. Delegated monitors, large and small: Germany's banking system 1800–1913. Journal of Economic Literature 40, 73–124.

Guinnane, T.W., 2003. A friend and adviser: external auditing and confidence in Germany's credit cooperatives, 1889–1914. Business History Review 77, 235–264.

Hilferding, R., 1910. Das Finanzkapital. Wiener Volksbuchhandlung, Wien.

Hoffmann, W.G., 1965. Das Wachstum Der Deutschen Wirtschaft Seit Der Mitte Des 19. Jahrhunderts. Springer Verlag, Berlin.

Hoffmann, W.G., Müller, J.H., 1959. Das Deutsche Volkseinkommen 1851–1959. J.C.B. Mohr, Tübingen.

Jeidels, O., 1905. Das Verhältnis Der Deutschen Großbanken Zur Industrie. Duncker & Humblot, Leipzig.

Kindleberger, C.P., 1993. A Financial History of Western Europe, second ed. Oxford University Press, Oxford.

King, R.G., Levine, R., 1993a. Finance and growth: Schumpeter might be right. Quarterly Journal of Economics 108, 717–737.

King, R.G., Levine, R., 1993b. Finance, entrepreneurship, and growth: theory and evidence. Journal of Monetary Economics 32, 513–542.

Komlos, J., 1978. The Kreditbanken and German growth: a postscript. Journal of Economic History 38, 476–479.

Levine, R., 1997. Financial development and economic growth: views and agenda. Journal of Economic Literature 35, 688–726.

Levine, R., Loayza, N., Beck, T., 2000. Financial intermediation and growth: causality and causes. Journal of Monetary Economics 46, 31–77.

Levine, R., Zervos, S., 1998. Stock markets, banks, and economic growth. American Economic Review 88, 537–558.

Luintel, K.B., Khan, M., 1999. A quantitative reassessment of the finance-growth nexus: evidence from a multivariate VAR. Journal of Development Economics 60, 381–405.

Neuburger, H., Stokes, H.H., 1974. German banks and German growth, 1883–1913: an empirical view. Journal of Economic History 34, 710–731.

Pagano, M., 1993. Finanical markets and growth: an overview. European Economic Review 37, 613–622.

Rajan, R.G., Zingales, L., 1998. Financial dependence and growth. American Economic Review 88, 559–586.

Reitmayer, M., 2002. Der Strukturwandel im Bankwesen und seine Folgen für die Geschäftstätigkeit der Privatbankiers im Deutschen Reich bis 1914. In: Pohl, H. (Ed.), Der Privatbankier. Nischenstrategien in Geschichte und Gegenwart. Bankhistorisches Archiv, Beiheft 41, Steiner Verlag, Stuttgart, pp. 11–26

Riesser, J., 1910. Die Deutschen Großbanken Und Ihre Konzentration. Verlag von Gustav Fischer, Jena.

Rousseau, P.L., Sylla, R., 2005. Emerging financial markets and early U.S. growth. Explorations in Economic History 42, 1–26.

Rousseau, P.L., Wachtel, P., 1998. Financial intermediation and economic performance: historical evidence from five industrialized countries. Journal of Money, Credit, and Banking 30, 657–678.

Sims, C.A., Stock, J.H., Watson, M.W., 1990. Inference in linear Time series models with some unit roots. Econometrica 58, 113–144.

Singh, A., 1997. Financial liberalisation, stock-markets and economic development. Economic Journal 107, 771–782.

Tilly, R.H., 1966. Financial Institutions and Industrialization in the Rhineland 1815–1870. University of Wisconsin Press, Madison.

Tilly, R.H., 1986. German banking, 1850–1913: development assistance for the strong. Journal of European Economic History 15, 113–152.

Toda, H.Y., Phillips, P.C.B., 1993. Vector autoregression and causality. Econometrica 61, 1367–1393.

Toda, H.Y., Yamamoto, T., 1995. Statistical inference in vector autoregressions with possibly integrated processes. Journal of Econometrics 66, 225–250.

Wixforth, H., Ziegler, D., 1994. The niche in the universal banking system: the role and significance of private bankers within German industry, 1900–1933. Financial History Review 1, 99–119.

Wysocki, J., 1993. Sparkassenorganisation und Wirtschaftswachstum von den Anfänge bis 1914. In: Mura, J. (Ed.), Sparkassenorganisation Und Wirtschaftswachstum: Historische Aspekte Und Zukunftsperspektiven. Sparkassen Verlag, Stuttgart, pp. 21–37.

Zapata, H.O., Rambaldi, A.N., 1997. Monte Carlo evidence on cointegration and causation. Oxford Bulletin of Economics and Statistics 59, 285–298.